# Gender Disparities in Canadian STEM Labor Force Participation:

# The Role of Family-Related Factors

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Lyncy Li

**Abstract:** This study investigates gender disparities in labor force participation (LFP) within Canada's STEM industries, focusing on the role of family-related factors such as marital status and childcare responsibilities. Using data from Statistics Canada's Labour Force Survey (2015), a logistic regression analysis is conducted in two stages. Stage 1 examines persistent gender disparities in STEM participation, while Stage 2 evaluates how family-related factors interact with gender to influence LFP. The findings reveal that women are significantly underrepresented in STEM and disproportionately affected by family care responsibilities, particularly the presence of young children, which negatively impacts their participation. These results underscore the need for targeted policies, including affordable childcare and flexible work arrangements, to reduce gender disparities in STEM and support equitable workforce participation. This study contributes to the literature by addressing the intersection of gender and family factors in the Canadian STEM context, providing insights for policymakers and organizations aiming to foster inclusivity and productivity in these critical fields.

## Introduction

Persistent gender disparities in labor force participation (LFP) in STEM fields remain a critical issue in Canada. Women face unique barriers, including wage gaps, career interruptions due to family responsibilities, and underrepresentation in high-demand sectors. These challenges, often intensified by structural and cultural factors, are well-documented in the literature. Despite these insights, few studies have examined the interaction between gender and family-related factors in the Canadian STEM context. This study fills that gap by investigating the role of family responsibilities in shaping women's participation in STEM.

Using data from Statistics Canada's Labour Force Survey (2015), we analyze these disparities in two stages. First, we assess whether gender differences in STEM LFP persist, using logistic regression with LFP as the outcome and gender as the primary predictor. In the second stage, we evaluate how family-related factors such as marital status and childcare interact with gender to influence LFP. This approach allows us to capture the specific challenges women face in balancing family responsibilities with STEM careers. Preliminary results suggest that women in STEM are significantly more likely than men to bear childcare responsibilities, which negatively impacts their labor force.

The findings will inform policies aimed at supporting women in the labor force. Unlike broad labor market interventions, these policies should focus on reducing the unique burdens on women, such as enhancing childcare access and promoting workplace flexibility. Targeted measures are essential for advancing gender equality in STEM, ensuring women have equitable opportunities to participate in these critical fields.

**Literature review**

Women's labor force participation has increased significantly over recent decades, yet globally women make up nearly half of the workforce but are underrepresented in fields such as science, technology, engineering, and mathematics (STEM), which have historically been male-dominated. Blau and Kahn's (2017) paper, "The Gender Wage Gap: Extent, Trends, and Explanations," provides a comprehensive analysis of the factors contributing to gender disparities in wages across different industries, including high-skill sectors like STEM.

Focusing on STEM is essential because it emphasizes meritocracy and talent, creating an environment where gender should theoretically play a minimal role. However, Glass (2013) figures out the underrepresentation of women in STEM suggests the presence of structural and societal barriers that uniquely disadvantage them. Following this idea, Kleven (2019) addresses factors such as family responsibilities, workplace culture, and implicit biases often intersect, disproportionately impacting women and contributing to challenges in recruitment, retention, and career progression.

By investigating women's participation in STEM, this research seeks to uncover the underlying dynamics in a sector critical to economic growth and emblematic of broader struggles in achieving gender equity. Understanding these nuanced factors is vital for developing policies that foster inclusivity and leverage the full potential of the workforce.

Female labor participation has long been of interest in labor economics. Historically women labor entered the market as economic scale expands and male labor absence. Hutchins(1916) explores the position of women workers after World War I, noting that the absence of male labor created expanded opportunities for women in various industries. However,

he also raised concerns about whether women's wages would be sustained as society stabilized and birth rates recovered. Butz and Ward (1979) argue high employment levels and wage variations among young women sustain countercyclical fertility trends, as seen in 1970 and 1974, with no expected change while employment remains strong. Female employment and wages are projected to rise with economic growth. Hill's (1983) studies typically consider economic and demographic characteristics such as female earnings, male earnings, non-earnings income, education, work experience, age, and number of children as determinants of labor supply for women.

Following this paper, economic researchers start various works focusing on family factors. Brinton et al., (1995) examined the case of South Korea and Taiwan, two societies that were similar in terms of female labor supply conditions and cultural values but have diverged in terms of married women's employment. The differences were attributed to changes in labor demand rather than supply. Fernandez (2004) found that women whose husbands had working mothers were more likely to participate in the labor force, suggesting that family background plays a role in shaping women's decisions to work. Ettner (1995) analyzed the impact of caregiving for disabled elderly parents on female labor supply and found that co-residence with a disabled parent led to a significant reduction in work hours, primarily due to withdrawal from the labor force. Bloom (2007) studied the effect of fertility on female labor force participation and found that legal restrictions on abortion can impact fertility rates, which in turn affect women's labor supply. Fernandez and Fogli (2007) explored the role of culture and learning in shaping female labor force participation. Fernandez (2007) developed a dynamic model of culture to understand the changing dynamics of women's participation in the workforce over time, while

Fogli (2008) argued that local information transmission and observation of nearby employed women can influence women's decisions to work.

For more specific area research, Agüero (2011) used infertility as an instrument to study the causal relationship between children and female labor force participation in developing countries, finding that the presence of children can bias estimates of women's labor force participation upwards. Mercedes (1974) finds that while developed countries often show an inverse relationship between women's labor force participation and fertility, this trend is weaker or absent in developing nations, highlighting the impact of regional economic, social, and cultural factors.

Overall, research on female labor force participation has considered a wide range of factors, including economic, demographic, cultural, and policy-related influences on women's decisions to enter or exit the workforce. As previously noted, studies have highlighted the importance of family background, caregiving responsibilities, fertility rates, and cultural norms in shaping women's labor force participation. Female workers often face greater pressures than men related to childbirth and family responsibilities, which may contribute to their disadvantages in traditional industries that demand high physical effort and offer inflexible work hours.

The internet industry's flexibility, remote work, and demand for digital skills have boosted female labor participation, especially in developing regions where it bypasses barriers in traditional sectors, offering better pay and growth opportunities. Meanwhile, tech firms require employer diversity. Azam (2011) examined the impact of the Indian job guarantee scheme on labor market outcomes and found that it had a positive effect on female labor force participation, stressing the gender dimension of the program's impact. However, female labor participation tends to be less favorable.  Jergins (2023) challenges the common assumption of a gender-equity

paradox in STEM fields. While female workers bring valuable skills and perspectives, they often face additional burdens from family and childbirth responsibilities. Similarly, Henry, Meango, and Mourifie (2024) emphasize the influence of Roy models and gender-specific challenges in STEM, demonstrating that factors such as a mother's educational background and the presence of female faculty heavily impact women's choices to enter STEM fields.

The referenced literature deeply explores the challenges faced by women in STEM fields, particularly regarding culture, education, and immigration factors. However, it tends to overlook the specific impacts of family and childcare responsibilities on female labor participation.

On one hand, much of the literature focuses on the U.S. tech industry, where women in STEM roles often enjoy higher incomes and work flexibility. Advances in technology and access to higher education have also led many high-earning women in Silicon Valley to opt for egg-freezing technology or a childfree lifestyle, distancing themselves from the traditional challenges faced by female labor. U.S. data can introduce significant complexities and biases, as it often intertwines minority labor groups with female labor, complicating analysis. As another major immigration destination, Canada offers a simpler population structure and a developing tech sector.

While existing research on women's participation in STEM extensively examines factors such as gender, family responsibilities, race, and culture, it often treats these factors in isolation. There is a notable gap in exploring the interactions between gender and other variables, which could reveal unique challenges faced by women. Additionally, some factors may specifically impact women in STEM, such as maternity-related career interruptions or gender biases in evaluations, but these are rarely studied in comparison to their effects on men. However, childbearing may no longer be the issue. Ahn (2002) analyzes OECD data (1970–1995), showing

a shift in the Total Fertility Rates (TFR) to Female Participation Rates (FPR) correlation from negative in the 1970s to positive by the late 1980s, explained through neoclassical fertility and labor supply models like Butz and Ward (1979). In this paper, we choose family factors only to evaluate its interactive influence. Therefore, we assume there may exist family factors after childbearing influencing women labor participation in STEM. We choose two main family factors marital status and age of youngest child to see: women labor hurts from getting married (before having a child) or taking childcare (after having a child).

This paper follows the research method idea from Kleven's (2019) study that uses interaction terms to analyze how gender and family responsibilities impact labor market outcomes. Given the project's focus on Canada's STEM industry and the influence of family-related factors, we choose regression analysis and to control for various socio-demographic factors. After studying the relationship between family factors and female participation, we hope to offer insights for local government and corporate policies to better support women in STEM amid an economic downturn.

**Methodology**

We wish to investigate whether there are gender disparities in Canadian STEM field. If it is true, do family-related factors contribute to the gender bias? All data are sourced from Statistics Canada Labour Force Survey (LFS) website (2015). It contains labor participation by sectors, and gender, marriage, childcare. We will append monthly LFS in 2015. The analysis is structured into two stages. In Stage 1 we conduct a logistic regression with LFP (labor force participation) as the binary dependent variable and gender as the single independent variable. Stage 2 tells whether family-related factors play a role in explaining the gender disparity. We define LFP as a binary variable where 1 indicates participation in STEM employment and 0 indicates non-participation. The model will be:

$$logit(LFPi) = \beta_0 + \beta_1(Gender_i) + \beta_2(family\ factors_j) + \beta_j(Gender \times family\ factors_j) + \epsilon i$$

Stage 1: Assess Persistent Gender Disparities in STEM LFP

Null Hypothesis (H0): There is no significant difference in LFP between men and women in Canadian STEM fields ($\beta_1=0$). Gender is a binary variable where 1 could represent "female" and 0 represents "male". $\beta_0$ is the intercept, and $\beta_1$ represents the effect of gender on the odds of participating in STEM. If estimator $\widetilde{\beta_1}$ is negative, it implies that the odds of women being in STEM are lower than those for me.

Stage 2: Evaluate the Role of Family-Related Factors in Gender Differences

Null Hypothesis (H0): Family-related factors do not contribute to gender differences in STEM LFP $\beta_j = 0$, where j denotes different categories of family factors including marital status (j=3) and age of youngest child (j=4, 5, 6)). All the independent variables are categorized. Marital Status could be only zero (single) or one (married). The data on the age of the youngest

child were categorized into three groups based on the Labor Force Survey variables: children aged 0–5, children aged 6–17, and children aged 18–24. These categories were created by coding the reported ages of children into binary indicators. Specifically, if the youngest child in the household was aged 0–5, the variable YoungAge_0_5 was coded as 1, and 0 otherwise. This approach statistically tests whether family-related factors like marital status or children have a significant interaction with gender, contributing to differences in STEM LFP.

**Results**

This study investigates persistent gender disparities in Canadian STEM labor force participation (LFP) and the influence of family-related factors. Using logistic regression analysis in two stages, key findings highlight the multifaceted nature of gender gaps in STEM fields.

Table 1: Descriptive Statistics

| Variable | Type | Mean | Min | Max | Observations | Missing |
|---|---|---|---|---|---|---|
| LFP | Binary | 0.0323 | 0 | 1 | 102,095 | 0 |
| SEX | Binary | 0.515 | 0 | 1 | 102,095 | 0 |
| YoungAge_0_5 | Binary | 0.101 | 0 | 1 | 102,095 | 0 |
| MAR | Binary | 0.606 | 0 | 1 | 102,095 | 0 |
| YoungAge_6_17 | Binary | 0.0971 | 0 | 1 | 102,095 | 0 |
| YoungAge_18_24 | Binary | 0.183 | 0 | 1 | 28,810 | 73,285 |

Note: YoungAge_18_24 data has significant missing values and was excluded from Stage 2 regression.

The descriptive statistics reveal that the average labor force participation rate in STEM fields is low (3.23%), with women constituting about half of the sample (51.5%). The presence of young children aged 0–5 is reported in 10.1% of the observations, while 9.7% have children aged 6–17. Marital status indicates that 60.6% of respondents are married. The variable YoungAge_18_24 has significant missing data, with observations available for only 28.2% of the sample.

The results from the initial logistic regression model indicate a significant gender disparity in STEM labor force participation. Women are less likely to participate in STEM fields than men, with an odds ratio of approximately 0.68, supporting the presence of a gender gap.

Table 2: Logistic Regression Results (Stage 1)

| Predictor | Coefficient | Std. Error | z-value | p-value |
|---|---|---|---|---|
| Intercept | -2.83191 | 0.05406 | -52.38 | <0.001 |
| SEX (Female) | -0.38667 | 0.03583 | -10.79 | <0.001 |

Table 3: Logistic Regression Results (Stage 2)

| Predictor | Coefficient | Std. Error | z-value | p-value |
|---|---|---|---|---|
| Intercept | 0.06875 | 0.13507 | 0.509 | 0.611 |
| SEX (Female) | -0.2645 | 0.08551 | -3.093 | 0.002 |
| YoungAge_0_5 | 0.90721 | 0.25212 | 3.598 | 0 |
| MAR (Married) | 0.43491 | 0.1745 | 2.492 | 0.013 |
| YoungAge_6_17 | 0.6965 | 0.25192 | 2.765 | 0.006 |
| SEX × YoungAge_0_5 | -0.3462 | 0.16205 | -2.137 | 0.033 |
| SEX × MAR | -0.0919 | 0.11053 | -0.832 | 0.406 |
| SEX × YoungAge_6_17 | -0.0713 | 0.15833 | -0.45 | 0.653 |

Both Stage 1 and Stage 2 confirm a significant gender disparity in STEM labor force participation. Women are less likely to participate in STEM compared to men, with negative coefficients for SEX in both stages. The presence of young children significantly increases the odds of STEM participation overall. However, the interaction term SEX × YoungAge_0_5 suggests that the positive effect of young children on labor force participation is smaller for women than for men, indicating that family care disproportionately affects women.

Having children aged 6–17 is positively associated with labor force participation, but the interaction SEX × YoungAge_6_17 is not significant, indicating no notable difference between men and women. Similarly, being married increases the odds of STEM participation, but the

interaction SEX × MAR is not significant, suggesting marital status does not differentially affect men and women.

The variable YoungAge_18_24 was excluded from Stage 2 due to significant missing data. Including this variable would have drastically reduced the effective sample size, limiting the model's reliability and generalizability.

## Conclusion

Women are significantly underrepresented in STEM fields, with lower odds of participation compared to men. Family care responsibilities, particularly the presence of young children, have a differential impact on women's labor force participation compared to men. Although having young children increases participation overall, this effect is less pronounced for women, highlighting family care as a critical factor contributing to the gender gap in STEM participation. Other factors, such as older children and marital status, show positive associations with labor force participation but do not exacerbate the gender gap. These patterns may reflect the influence of older, more established laborers in STEM fields.

The government has a vital role in addressing these disparities and supporting women in STEM fields. This can be achieved through policies such as expanding access to affordable, high-quality childcare services, which would reduce the burden of family care responsibilities disproportionately carried by women. Additionally, the government should promote flexible work arrangements, including remote work options and adaptable schedules, to enable parents to balance family responsibilities with professional obligations. Implementing gender-sensitive policies in recruitment, retention, and promotion processes within STEM industries is also essential to address systemic biases and foster an equitable workplace environment. These measures, combined with public awareness campaigns and funding initiatives to encourage women's participation in STEM, can help reduce the structural and family-related barriers that perpetuate gender disparities in STEM labor force participation.

This study has several limitations that should be acknowledged. First, the analysis relies on cross-sectional data from Statistics Canada's Labour Force Survey (2015), which limits the ability to infer causal relationships or observe changes over time. The data also predates recent

labor market shifts, including those triggered by the COVID-19 pandemic, potentially affecting the generalizability of findings to current STEM dynamics. Additionally, while the focus on family-related factors such as marital status and childcare responsibilities provides valuable insights, the exclusion of other critical influences—such as workplace culture, mentorship opportunities, and regional variations—restricts the study's scope. Missing data on older children (ages 18–24) further limits the robustness of findings related to this demographic.

The logistic regression model employed simplifies the complex interactions between gender, family responsibilities, and labor force participation. Unobserved variables or omitted confounders may introduce biases, and the findings are specific to Canada, potentially reducing applicability to other countries. Future research could address these limitations by incorporating longitudinal data, exploring additional variables like employer policies, and conducting comparative analyses across different national contexts to provide a more comprehensive understanding of gender disparities in STEM.

For further research, it is crucial to examine the impact of the COVID-19 pandemic on female labor force participation in STEM fields. The data analyzed in this study are from 2015, before the pandemic, a period when family care responsibilities already disproportionately affected women. The pandemic exacerbated these challenges, with widespread school closures and increased childcare burdens further straining women's ability to participate in the workforce. However, the pandemic also accelerated the adoption of remote work, which has the potential to alleviate some of the pressures of family responsibilities by allowing more flexible work arrangements. This shift might have fundamentally altered the dynamics of female labor participation in STEM. Exploring post-pandemic labor market trends can provide valuable insights into whether remote work has mitigated the gender gap or if additional challenges have

emerged. Understanding how these changes have reshaped women's experiences in STEM will be critical for designing policies that address the evolving needs of the workforce.

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**Appendix**

Table 1: Descriptive Statistics

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| YoungAge_18_24 | Binary | 0.183 | 0 | 1 | 28,810 | 73,285 |

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| SEX (Female) | -0.38667 | 0.03583 | -10.79 | <0.001 |

Table 3: Logistic Regression Results (Stage 2)

| Predictor | Coefficient | Std. Error | z-value | p-value |
|---|---|---|---|---|
| Intercept | 0.06875 | 0.13507 | 0.509 | 0.611 |
| SEX (Female) | -0.2645 | 0.08551 | -3.093 | 0.002 |
| YoungAge_0_5 | 0.90721 | 0.25212 | 3.598 | 0 |
| MAR (Married) | 0.43491 | 0.1745 | 2.492 | 0.013 |
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| SEX × MAR | -0.0919 | 0.11053 | -0.832 | 0.406 |
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R code:

```
## ECON 472
install.packages("dplyr")
library(dplyr)
install.packages("skimr")
library(skimr)
SEX <- ifelse(table$SEX == 1, 0,
        ifelse(table$SEX == 2, 1, table$SEX))
MAR <- ifelse(table$MARSTAT == 1, 1,
        ifelse(table$MARSTAT == 3, 0,
          ifelse(table$MARSTAT == 2, 1,
            ifelse(table$MARSTAT == 4, 0,
```

```
                    ifelse(table$MARSTAT == 5, 0,
                        ifelse(table$MARSTAT == 6, 0,
                table$MARSTAT))))))
YoungAge_0_5 <- ifelse(table$AGYOWNKN %in% c(1, 2), 1, 0)
YoungAge_6_17 <- ifelse(table$AGYOWNKN %in% c(3, 5), 1, 0)
YoungAge_18_24 <- ifelse(table$AGYOWNKN == 6, 1, 0)
LFP <- ifelse(table$LFSSTAT == 1 & table$NAICS_18 == 11, 1, 0
        )
skim(SEX)
skim(MAR)
skim(LFP)
skim(YoungAge_0_5)
skim(YoungAge_6_17)
skim(YoungAge_18_24) # Missing var

model_stage1 <- glm(LFP ~ SEX, data = table, family = binomial(link = "logit"))
summary(model_stage1)
model_stage2 <- glm(LFP ~ SEX * YoungAge_0_5 + SEX * MAR + SEX * YoungAge_3_5
            + SEX * YoungAge_6_12,+ SEX * YoungAge_13_17 + SEX * YoungAge_18_24,
            data = table, family = binomial(link = "logit"))

# Summary of the model
summary(model_stage2) # Invalid
```

```
# Weight all factor
table$LFP <- ifelse(table$LFSSTAT == 1 & table$NAICS_18 == 11, 1, 0)
weights <- ifelse(table$LFP == 1,
                  1,
                  sum(table$LFP == 1) / sum(table$LFP == 0))
length(weights)
sum(is.na(table$LFP))  # Count missing values in LFP
sum(is.na(weights))    # Count missing values in weights

model_stage2_weighted <- glm(LFP ~ SEX * (YoungAge_0_5 + MAR + YoungAge_6_17),
                             data = table, family = binomial(link = "logit"),
                             weights = weights)
summary(model_stage2_weighted)
```